Sixdof PX4 Integration Instructions

Introduction

In this document we will describe the process of integrating the Sixdof Sensor with a PX4 drone controller. The following instructions use the Holybro X500 V2 development kit drone, with the Sixdof Sensor, a 32bit Raspberry Pi 3b, and a Linux computer running QGroundContol Software. Additionally, in this experiment a Xbox joystick was used to control the drone, alternatively a full drone remote control can be used as well.

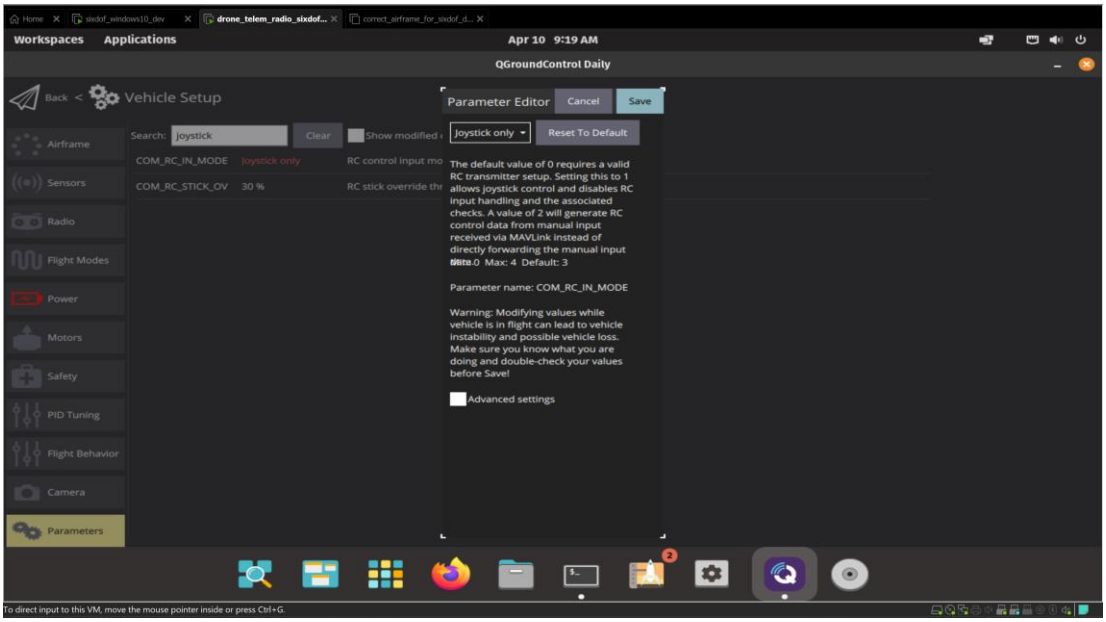
QGroundControl

The following instructions describe how to set up QGroundControl software:

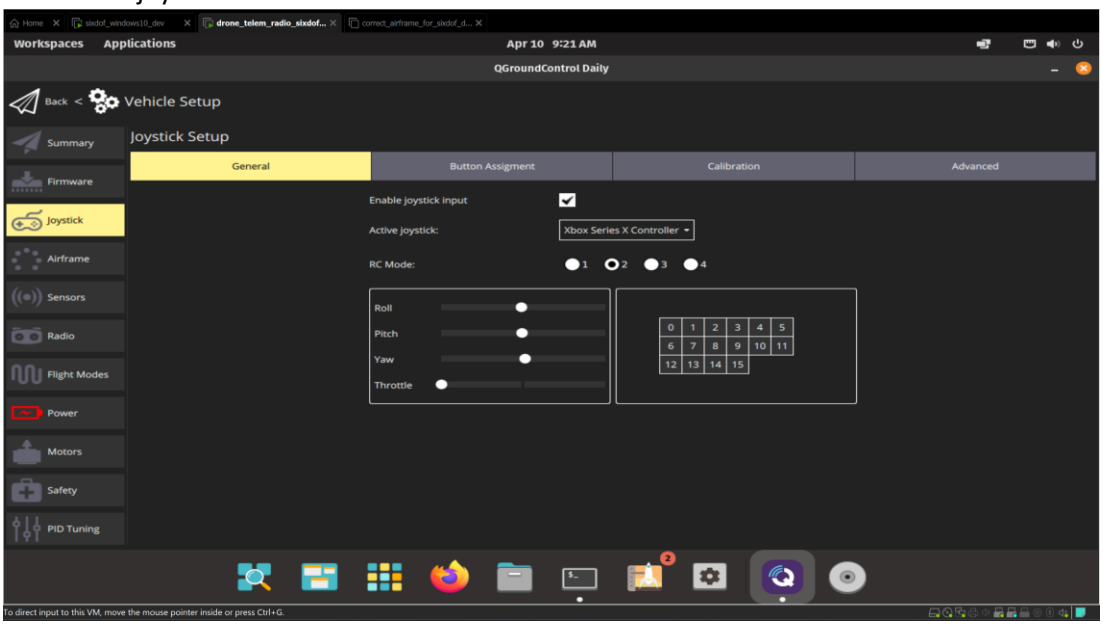
1. Download the QGroundControl daily build of the master branch
    a. The master branch was selected instead of the release because in the stable release our specific drone model (Holybro X500) isn't one of the options.
2. Plug-in the Holybro USB TELEMETRY dongle into the USB port of the computer.
3. Make sure that the battery of the drone is connected, and the propellers are detached.
4. Plug-in the Xbox Controller into the USB port of the computer.
5. Plug-in the drone's battery.
6. Choose the correct drone model:



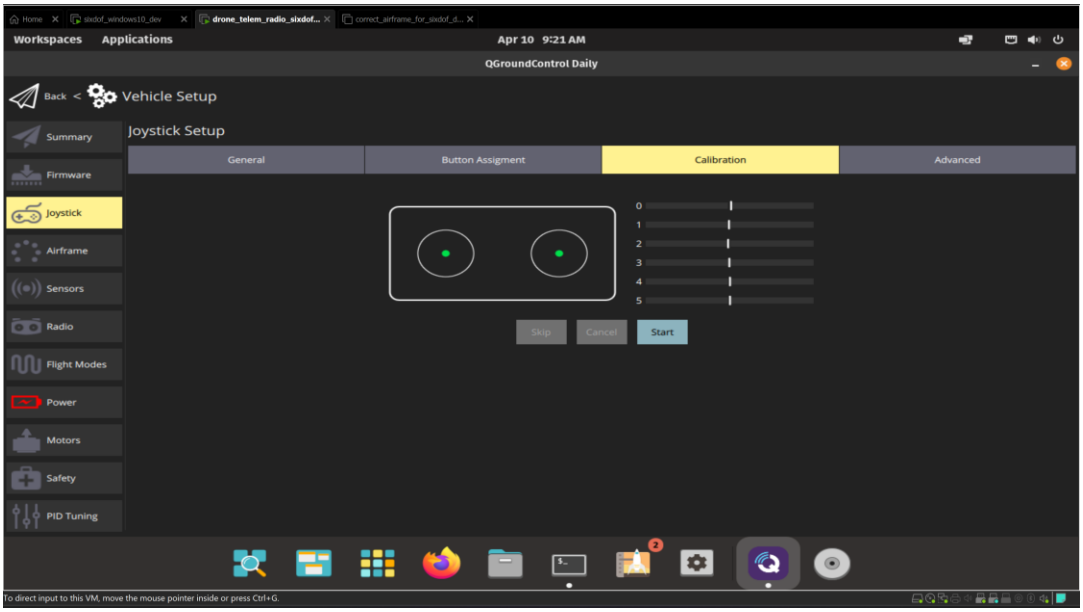
7. Change to “Joystick only” so that the drone controller will allow you to control it using the Xbox controller



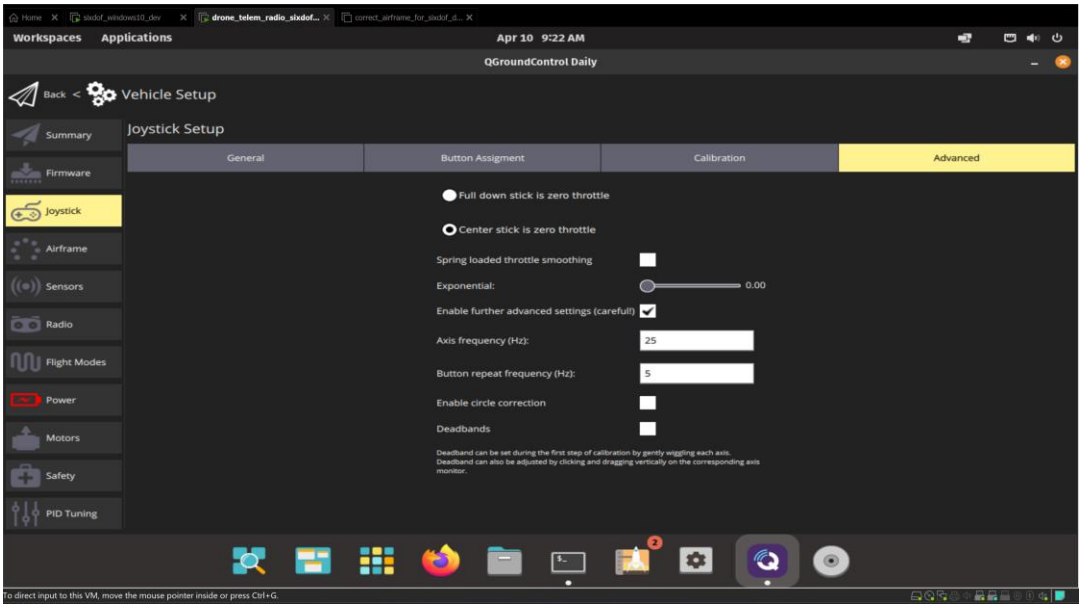
8. Enable the joystick



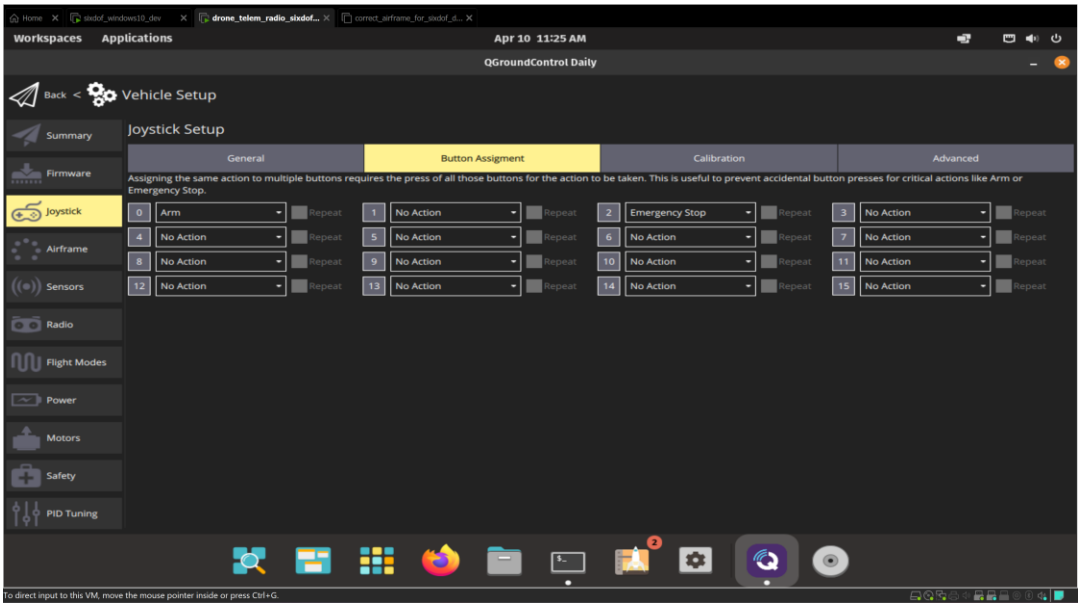
9. Calibrate the joystick



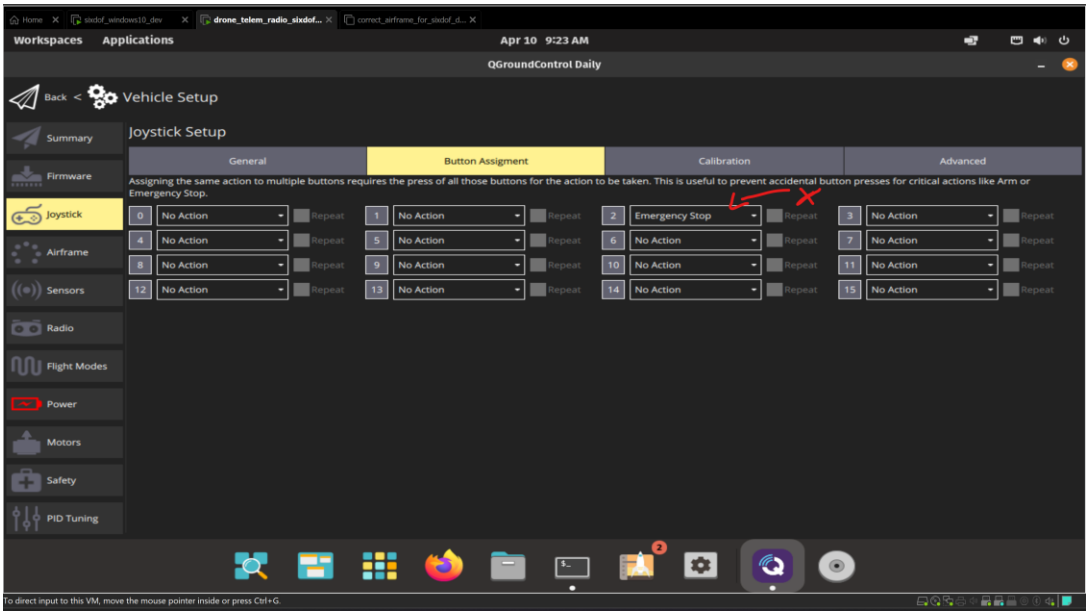
10. Make the drone safer to control by having the drone's motors go to minimum speed when letting go of the joystick.



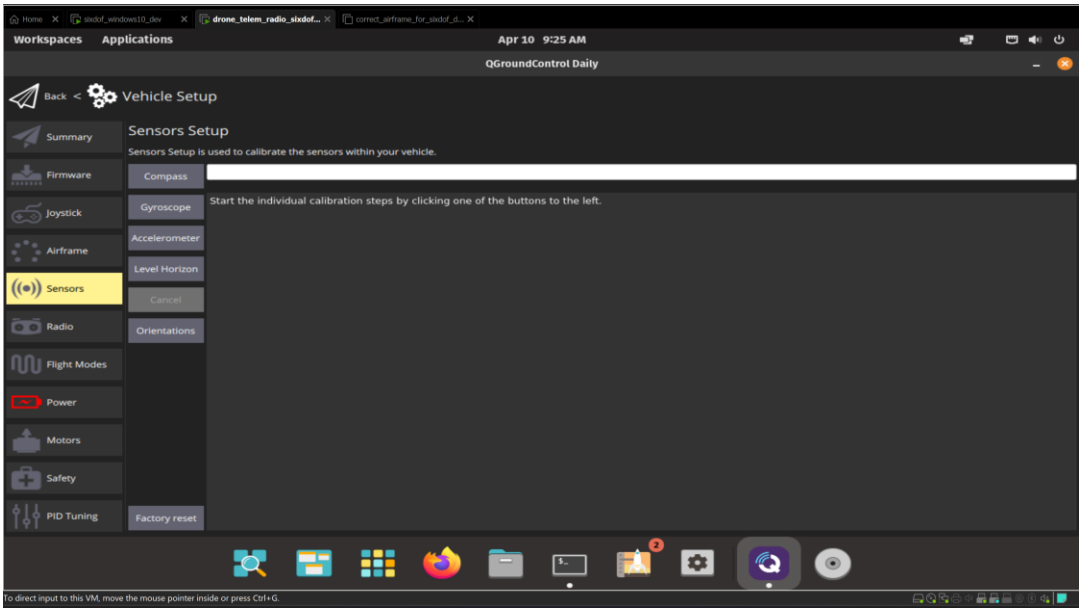
11. Set the “A” button on the Xbox Controller to arm the drone



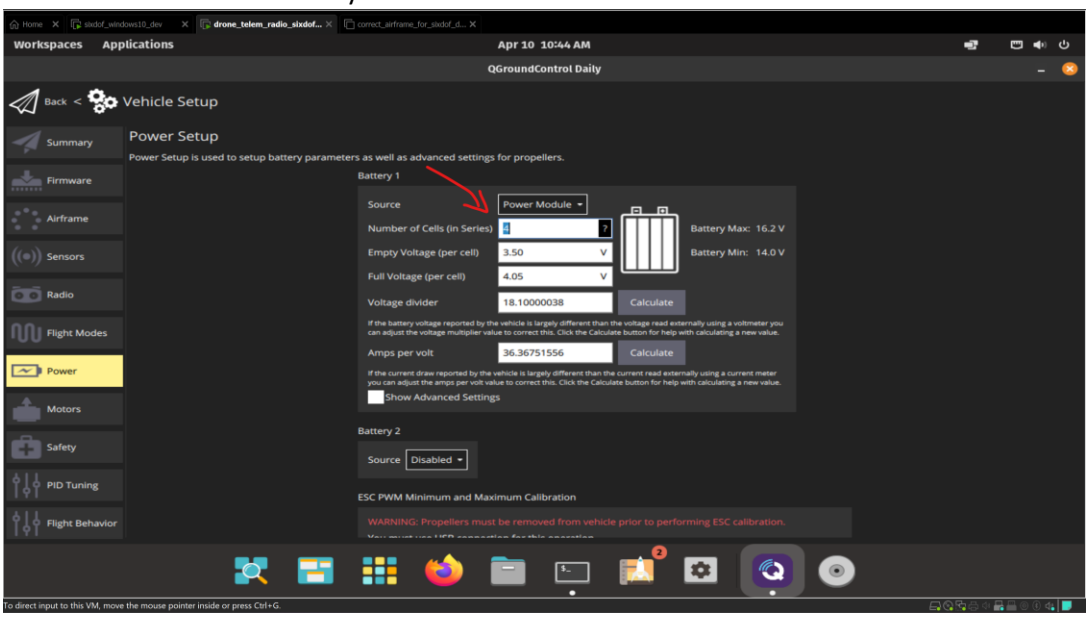
12. Set the “X” button on the Xbox Controller to kill the drone (disarm while flying).



13. Calibrate the compass.

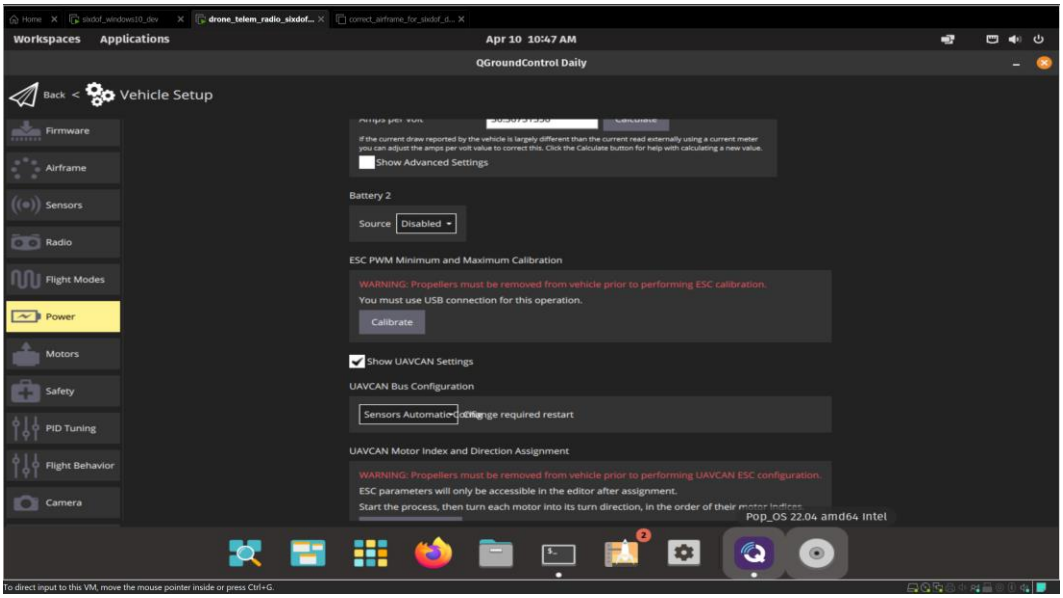


14. Set correct number of battery cells

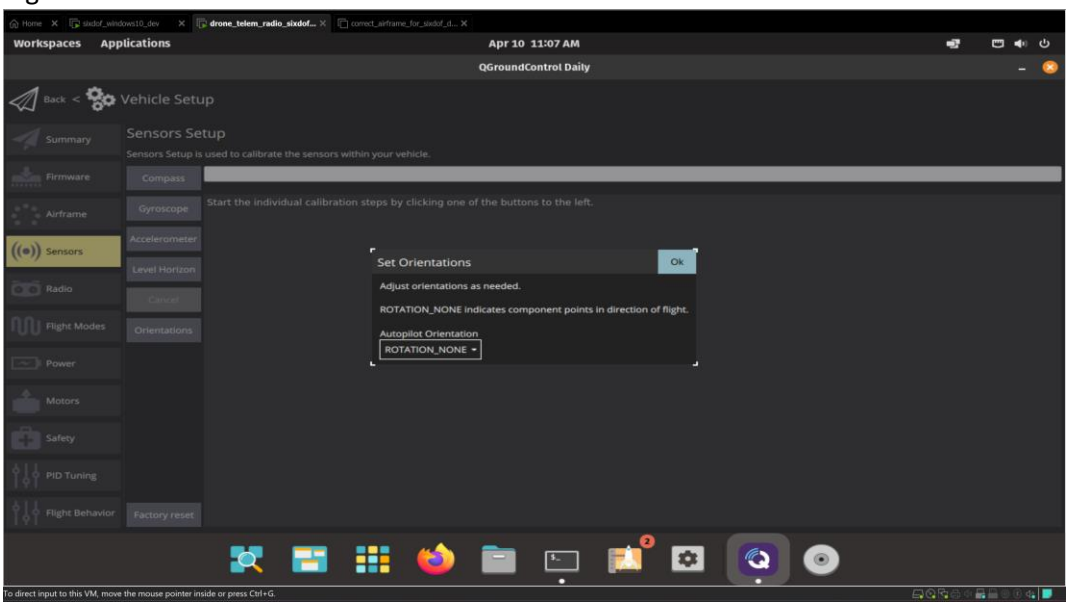


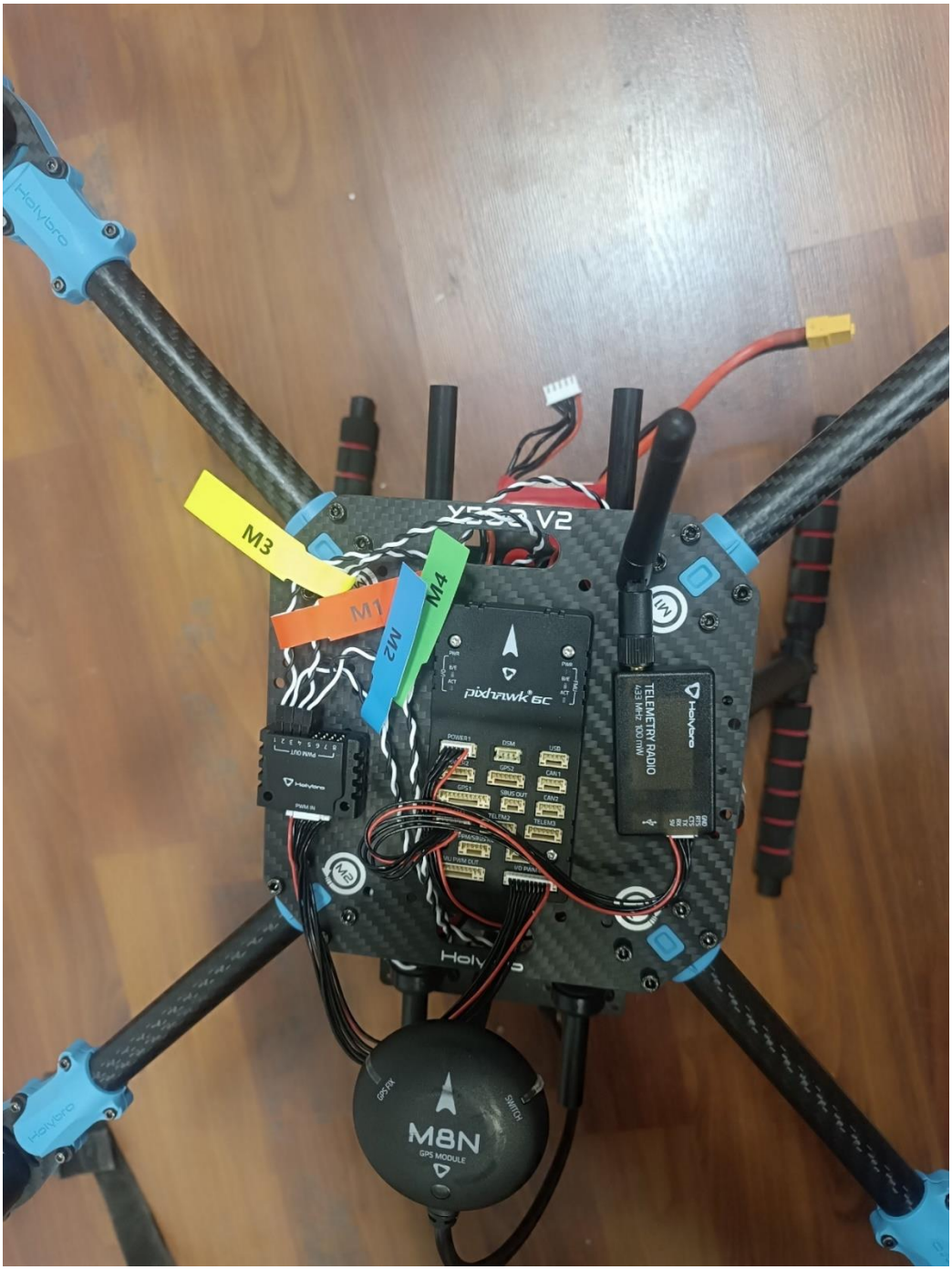
15. Detach the battery and plug in the Pixhawk 6C's USB C port into the laptop's USB port.

16. Calibrate ESC PWM

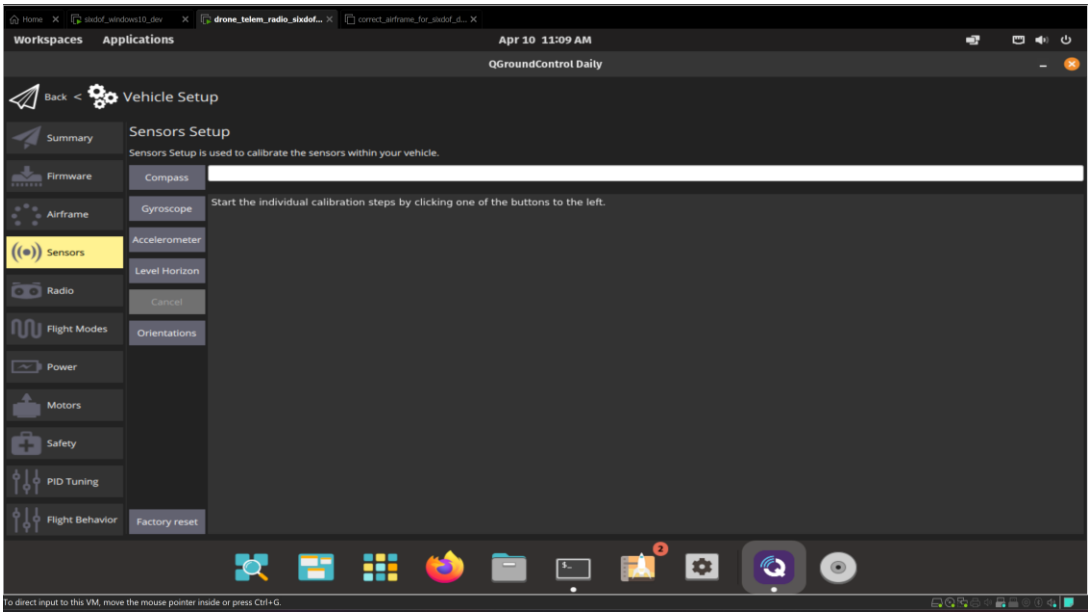


17. Disconnect the battery again (the motor calibration requires connecting the batteries).
18. Disconnect the drone controller from USB C and reconnect the battery so that it can be controlled (again) via the telemetry radio.
19. Make sure that the drone controller board is physically pointing towards the direction of flight

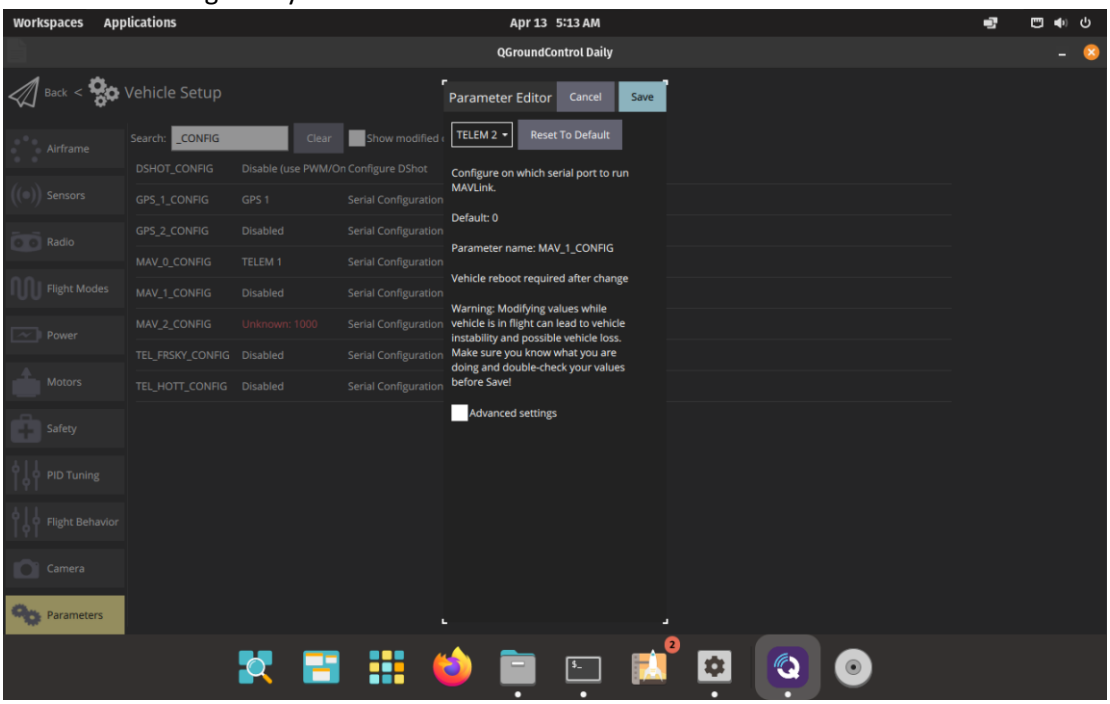



M3
M1
M2
M4
pixhawk 6C
TELEMETRY RADIO
M8N
GPS MODULE

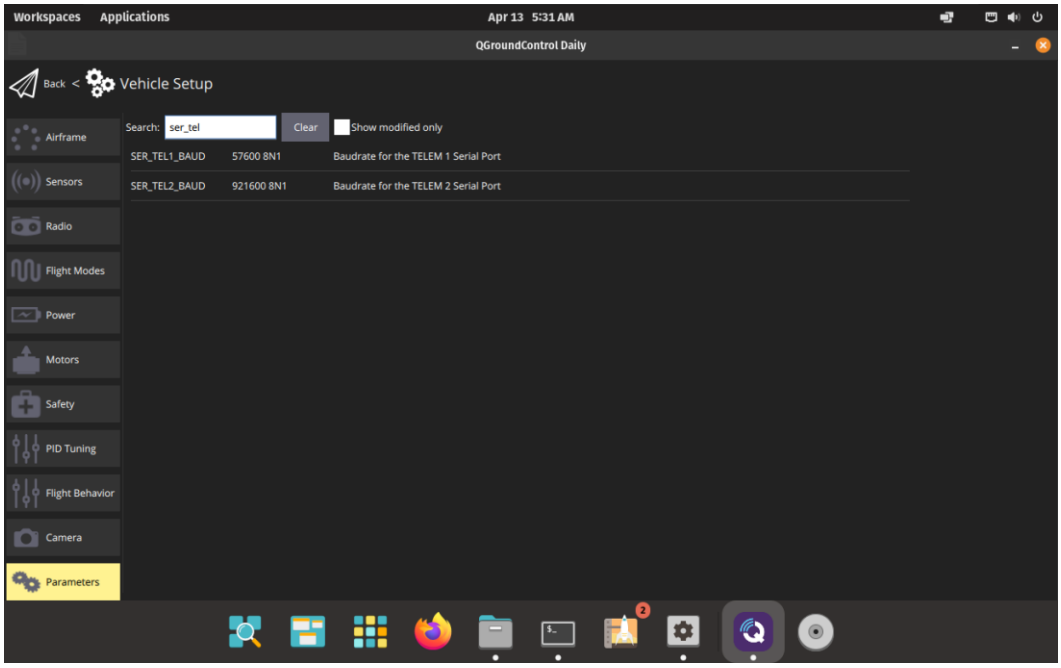
20. Calibrate gyro, accelerometer, level horizon



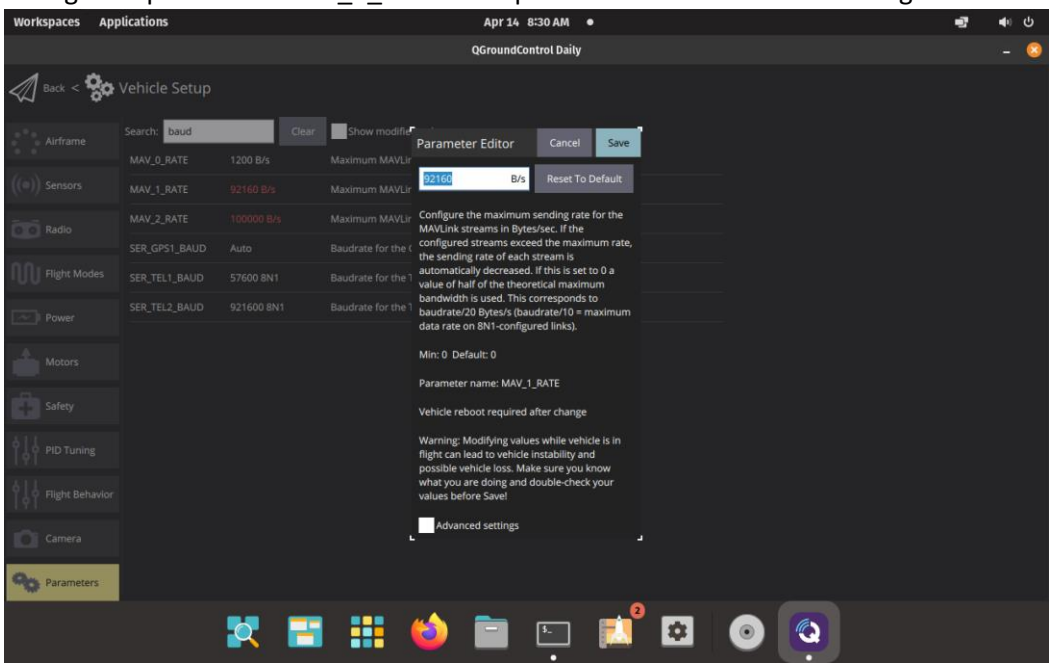
21. Configure the serial port TELEM2 to be activated in mavlink mode, to communicate with the Raspberry Pi. TELEM1 is configured for MavLink mode by default (baud rate 57600), but TELEM2 is unassigned by default.



22. Leave the TELEM1 and TELEM2 serial port communication baud rate values the default.

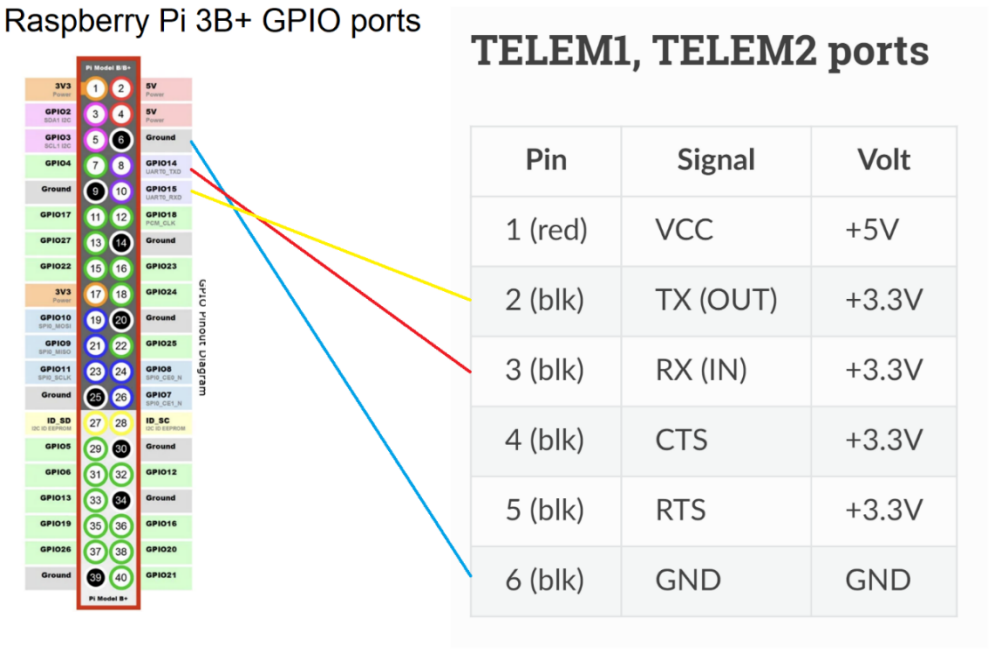


23. Change the parameter MAV_1_RATE to a specific number that should be a good balance:



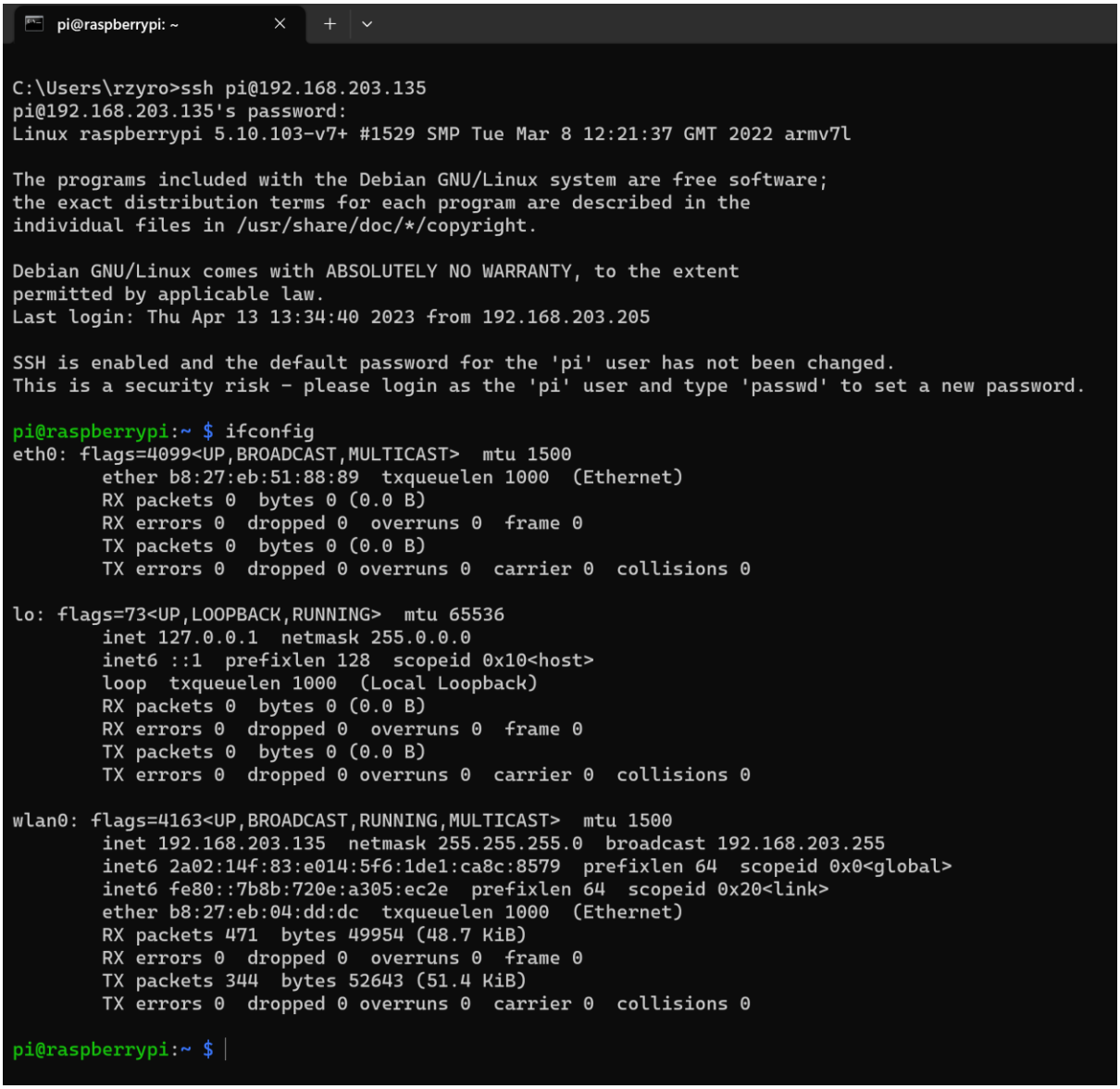
24. Now disconnect the drone's battery, make sure that the telemetry radio is connected to TELEM1, and connect the raspberry pi to TELEM1. Connect "ground- GPIO pin 6 of the rpi" to "ground- pin in the TELEM2 port". Connect "tx- GPIO pin 8 of the rpi UART0_TXD a.k.a GPIO14" to the "rx- in the TELEM2 port". Connect "rx- GPIO pin 10 of the rpi UART0_RXD

a.k.a GPIO15” to the “tx- in the TELEM2 port”.



25. Don’t connect the 5v pin of the Raspberry Pi to the 5v pin of the TELEM2 port because the Raspberry Pi uses up to 2.5 amps of power, the TELEM1 port is only rated to supply up to 1.5 amps of power. Instead use a battery pack that supports 2.5 amps.
26. Connect the battery of the done.
27. View the raspberry pi via HDMI on a computer monitor while it’s mounted on the drone, so that we can set up SSH.
28. Run the command: sudo raspi-config
29. Choose: “System options” -> “Wireless LAN” and enter your wifi network’s name and password.
30. Go back to the main menu of raspi-config and select “Interface options” -> “Serial Port”: Click ENTER.
31. Set “Would you like a login shell to be accessible over serial?” to “no”.
32. Set “Would you like serial port hardware to be enabled?” to “yes”.
33. Do “Finish” and then click “yes” on “Would you like to reboot now?”.
34. Now everything is configured so that in the code running on the raspberry pi, that uses the mavsdk library, use the address: "serial:///dev/serial0:921600".
35. Run “sudo apt update” and “sudo apt upgrade” on the rpi.
36. Setup ssh on the raspberry pi. Run sudo raspi-config and nagivate to: “Interface Options” -> “P2 SSH”. Click ENTER and choose Yes.
37. Reboot the rpi.
38. You can then this the CMD windows command to scan for all IPs that are connected to the same hotspot as your laptop, replace 192.168.203 with the subnet of your WiFi hotspot which can be found by running ipconfig on the windows laptop.
    for /L %i in (1,1,255) do @ping -n 1 -w 100 192.168.203.%i | find "TTL="
39. Now connect a laptop to the same WiFi network as the Raspberry Pi is connected to, and control it using SSH command. If you don’t have ssh, you can download and

install “Putty” for Windows.



40. Various versions:

```
pi@raspberrypi:~ $ lscpu
Architecture:            armv7l
Byte Order:              Little Endian
CPU(s):                  4
On-line CPU(s) list:     0-3
Thread(s) per core:      1
Core(s) per socket:      4
Socket(s):               1
Vendor ID:               ARM
Model:                   4
Model name:              Cortex-A53
Stepping:                r0p4
CPU max MHz:             1400.0000
CPU min MHz:             600.0000
BogoMIPS:                38.40
Flags:                   half thumb fastmult vfp edsp neon vfpv3 tls vfpv4 idiva idivt vfpd32 lpae evtstrm crc32
pi@raspberrypi:~ $
```

```
pi@raspberrypi: ~
pi@raspberrypi:~ $ uname -a
Linux raspberrypi 5.10.103-v7+ #1529 SMP Tue Mar 8 12:21:37 GMT 2022 armv7l GNU/Linux
pi@raspberrypi:~ $ ldd --version
ldd (Debian GLIBC 2.28-10+rpt2+rpi1+deb10u1) 2.28
Copyright (C) 2018 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
Written by Roland McGrath and Ulrich Drepper.
pi@raspberrypi:~ $ g++ --version
g++ (Raspbian 8.3.0-6+rpi1) 8.3.0
Copyright (C) 2018 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

pi@raspberrypi:~ $ make --version
GNU Make 4.2.1
Built for arm-unknown-linux-gnueabihf
Copyright (C) 1988-2016 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
```

41. Run this command to stop the serial console interface (that might interfere with MavLink on the /dev/serial0 port), this step is probably not needed since raspi-config should have done this for us: sudo systemctl stop serial-getty@ttyS0.service && sudo systemctl disable serial-getty@ttyS0.service